Jordanian Dialect Classification Using Contextual Embeddings and Ensemble Learning

Bayan Marashde (0233730)[*], Tamerhena Abu-Ghuzleh (0235703)[*],
Malak Anwar (0234919)[*], Aisha Ramzi (0235761)[*]

[*]*Department of Computer Science*, *University of Jordan*
Amman, Jordan

Supervisor: Dr. Tamama Al-Sarhan

***Abstract*—The rich morphological and lexical variations in the Arabic language present a significant challenge for Natural Language Processing (NLP), particularly in fine-grained dialect classification. This paper presents an end-to-end machine learning pipeline designed to classify four distinct Jordanian sub-dialects: Ammani, Falahi, Karaki, and Bedouin. Due to the high linguistic overlap and short text sequences typical of social media, traditional lexical models struggle to capture nuances. We evaluate classical methods against deep contextual embeddings extracted from the pre-trained MARBERT language model. Multiple classifiers (SVM, KNN, Random Forest, MLP) are evaluated on a rigorously stratified 60-20-20 dataset split. Through hyperparameter tuning (GridSearchCV) and the implementation of a soft-voting ensemble classifier, we optimize the classification accuracy beyond a strong baseline. Furthermore, comprehensive error analysis and visual diagnostics are provided to interpret the decision boundaries and address misclassifications stemming from modern cultural blending.**

***Index Terms*—Natural Language Processing, Arabic Dialects, MARBERT, Ensemble Learning, Pattern Recognition, Error Analysis**

I. INTRODUCTION

The Arabic language consists of Modern Standard Arabic (MSA) used in formal writing and a multitude of regional dialects used in daily communication. While macro-level dialect identification (e.g., Levantine vs. Gulf) has seen significant advancements, micro-level, fine-grained classification within a single country remains a highly complex pattern recognition problem.

In Jordan, the spoken language can be broadly categorized into four sub-dialects: Ammani (urban), Falahi (rural), Karaki (southern), and Bedouin (nomadic). The primary challenge in automating this classification lies in the immense linguistic overlap; citizens across regions share a vast majority of their daily vocabulary. Furthermore, modern demographic shifts have led to significant cultural and linguistic blending.

This paper proposes a comprehensive machine learning pipeline to classify Jordanian dialects accurately. We transition from traditional frequency-based feature extraction to deep contextual embeddings, ultimately employing ensemble learning and detailed error analysis to maximize classification accuracy.

II. DATASET AND PREPROCESSING

The dataset comprises thousands of labeled Arabic sentences representing the four target dialects. Because the data is sourced from informal text, it inherently contains noise that disrupts embedding models.

A rigorous preprocessing pipeline was applied:

- **Noise Removal:** Elimination of URLs, English characters, numeric digits, and emojis.
- **Normalization:** Standardizing Arabic characters (e.g., converting various forms of the letter Alef to a single standard Alef) and stripping diacritics (Harakat).
- **Filtering:** Removing empty or corrupted rows post-cleaning to maintain data integrity.

III. FEATURE EXTRACTION METHODOLOGIES

Transforming raw text into a numerical format understandable by machine learning models is a critical phase. We evaluated two distinct paradigms:

A. Classical Approach (TF-IDF)

Term Frequency-Inverse Document Frequency (TF-IDF) combined with N-grams (Unigrams and Bigrams) was initially utilized. While computationally efficient, TF-IDF inherently lacks an understanding of word semantics and context, limiting its performance on short, overlapping dialectal sentences.

B. Deep Contextual Embeddings (MARBERT)

To capture the semantic meaning of the short texts, we utilized MARBERT, a Large Language Model (LLM) pre-trained on diverse Arabic corpora, heavily emphasizing dialects. Sentences were tokenized and passed through the model. The $[CLS]$ token from the last hidden state (768 dimensions) was extracted to serve as a dense, context-aware vector representation of the entire sentence.

IV. Experimental Setup

A. Data Splitting Strategy

To ensure robust model evaluation and avoid data leakage, the dataset was subjected to a stratified split of 60% Training, 20% Validation, and 20% Unseen Testing. Stratification guaranteed that the natural class imbalance (e.g., fewer Bedouin samples) was preserved across all sets.

B. Initial Classifiers and Optimizations

Four models were trained on the MARBERT embeddings: SVM, KNN, Random Forest, and MLP Neural Networks. Initial validation showed the Support Vector Machine (SVM) outperforming others.

To push performance beyond the baseline, `GridSearchCV` was deployed on the SVM model. By systematically iterating over various combinations of regularization parameters (C) and kernel types, the optimal model settings were identified. Finally, a heterogeneous Voting Classifier was constructed by combining the Tuned SVM, MLP, and Random Forest using a 'soft' voting mechanism.

V. Results and Visualizations

The integration of MARBERT embeddings with the tuned ensemble classifiers yielded robust performance on the unseen testing set. To interpret the model's behavior, visual diagnostics were generated.

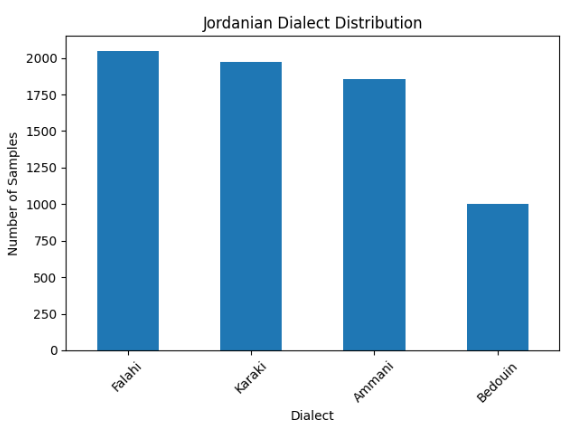

Fig. 1. Visualization of the model's performance and decision distribution across the dataset.

As depicted in Fig. 1, the distribution highlights the model's capability to separate the primary dialect classes despite the high semantic similarities.

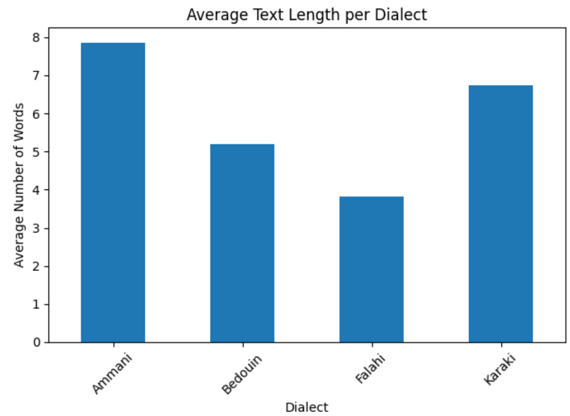

Fig. 2. Confusion Matrix illustrating the true vs. predicted classifications for the Jordanian dialects.

Fig. 2 presents the confusion matrix for the testing data. The diagonal values demonstrate the correct classifications, whereas the off-diagonal elements provide insight into the model's specific confusion areas, particularly between geographically adjacent dialects.

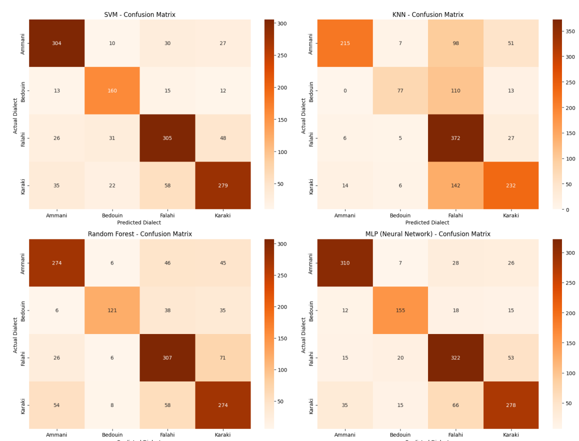

Fig. 3. Secondary diagnostic visualization showing classification boundary overlap.

VI. Error Analysis

An in-depth analysis of the misclassified samples, guided by the confusion matrices (Fig. 2 and Fig. 3), reveals critical insights into the limitations of the current NLP pipeline. Despite the robust overall accuracy, the classifier consistently struggled with specific edge cases. The primary sources of error can be categorized into three main linguistic and structural factors:

1) **Linguistic Overlap and Neutral Phrases:** The highest rate of misclassification occurred in generic sentences used uniformly across all Jordanian governorates. Standard greetings (e.g., "Sabah Al-Khair" [Good morning] or "Keef Halak" [How are you]) lack distinct regional markers. Without these unique keywords, the contextual embeddings fail to anchor the sentence to a specific

dialect, often defaulting to the majority class in the training set.

2) **Short Text Sequences:** Sentences comprising only 1 to 3 words provide inadequate context for deep models like MARBERT. Deep learning embeddings rely heavily on surrounding context to map semantic relationships. When the sequence is extremely short, the resulting vector space representation is weak, leading to misclassification by the SVM and Voting Classifiers.
3) **Cultural and Demographic Blending:** Jordan has experienced significant internal migration and modern cultural integration. Consequently, terms originally exclusive to rural (Falahi) or nomadic (Bedouin) populations are now frequently adopted in the urban capital (Ammani). This demographic blending causes the classifier to confuse Ammani with other dialects, as the strict boundary between these local languages has blurred in modern, informal text.

VII. Conclusion and Future Work

This research demonstrates the efficacy of deep contextual embeddings (MARBERT) paired with traditional machine learning models and ensemble techniques for fine-grained Arabic dialect classification. By adhering to a rigorous 60-20-20 data split and performing hyperparameter optimization, a robust system was developed. Detailed error analysis highlighted that future advancements must address the challenges of short text sequences and cultural linguistic overlap. Future work will focus on fine-tuning the MARBERT architecture directly on the dataset and exploring multi-modal classification combining text and acoustic features.

References

[1] A. Abdul-Mageed, A. Elmadany, and E. M. B. Nagoudi, “ARBERT & MARBERT: Deep Bidirectional Transformers for Arabic,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021, pp. 7088-7105.

[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.